

Advance Java Flow

- 3-Tire application working
 - What concepts will be covered in upcoming classes (Overview)
- Project with File Handling CRUD Operations
 - Why we cannot use collection framework to store data
 - Problems over file
 - Why we need Database

- JDBC
 - Why we need Database
 - JDBC API Introduction
 - What is jar file
 - Ways to add jar file in java application
 - Working of JDBC
 - Coupling
 - Tight coupling
 - Loose Coupling
 - JDBC 5 Steps
 - Load and Register Driver Class
 - Ways to load
 - Establish Connection between java application and database server
 - Url,userName>Password
 - Ways to establish connection
 - Create platform to execute queries
 - Ways to create platform
 - Statement
 - PreparedStatement
 - What is placeholder and rules to use it
 - Execute Query
 - Ways to execute queries
 - ResultSet
 - Close costly resources
 - Ways to close costly resources
 - Batch Execution
- Servlet
 - Server
 - What is server
 - Types of Server
 - Working of server
 - Request and response handling
 - Servlet
 - What is servlet

- Ways to create servlet
 - Generic Servlet
 - HttpServlet
- Steps to Create Dynamic web Project
 - Ways to add deployment descriptor
- Servlet Creation using Generic Servlet
- Servlet Mapping
 - web.xml
 - Annotation
- Fetching user data from request
- Servlet Creation using HttpServlet
 - Methods of HttpServlet
- Servlet Chaining
 - RequestDispatcher
 - sendRedirect
- Passing data from one Servlet to another
 - RequestObject
 - HiddenFormFields
 - QueryString
 - SessionObject
 - Methods of session
- Exception and Error handling in servlet
- Servlet life cycle
- Load on startup
 - Using xml
 - Using annotation

Project using JDBC + SERVLET

- Jsp(Java Server page)
 - What is jsp
 - Why jsp
 - Tags in jsp
 - Scriptlet Tag
 - Declaration Tag
 - Directive Tag
 - Expression Tag
 - CommentTag
 - Implicit Objects in jsp
 - Exception and Error handling in JSP
 - JSP LifeCycle

Project using JSP + JDBC

- Hibernate
 - Maven Project Creation
 - Maven build tool
 - pom.xml
 - Maven Responsibilities
 - What is a Framework?
 - Problems with JDBC
 - Singleton Design Pattern
 - What is hibernate?
 - What is ORM?
 - What is persistence.xml
 - Dialect
 - What is JPA
 - Advantages Of Hibernate
 - Cache memory
 - Factory Design Pattern
 - Layered Packaging in project
 - Hibernate architecture
 - Javax.persistence api (Classes and Interfaces)
 - Hibernate Mapping (UNI & BI Directional)
 - @OneToOne
 - @OneToMany
 - @ManyToOne
 - @ManyToMany
 - Composite Key
 - Annotations in Hibernate
 - Hibernate Entity Life Cycle

Project using Jsp/Servlet + Hibernate

- Spring
 - History of Spring
 - Why it is named as Spring
 - Why Spring framework
 - Advantages of Spring Framework
 - Spring Container/IOC Container
 - BeanFactory
 - ApplicationContext
 - Bean object configuration and dependency injection
 - XML configuration
 - Constructor Injection
 - Setter Injection
 - Java Class configuration

- Has A Relationship
 - XML configuration
 - Constructor Injection
 - Setter Injection
 - Java Class configuration
- List Injection
- Only one xml configuration
- No xml configuration / annotation configuration
 - Has a relationship
- SpringBoot
 - Application development issues in exchanging data
 - Why API Instead of Direct DB Access?
 - Introduction to JSON Objects
 - Json Object Creation for
 - Java class
 - Java class with List
 - Has a relationship (Nested Objects)
 - What is SpringBoot?
 - Key Features of Spring Boot
 - Advantages of Spring Boot
 - Ways to create the Spring Projects
 - Using Spring initializr (start.spring.io)
 - STS(Spring Tool Suite)
 - *Layers of Spring Boot Architecture*
 - Controller Layer
 - Service Layer
 - Data Access Object Layer
 - Repository Layer
 - Data Transfer Object Layer

Project using SpringBoot CRUD Operations Using API

- ***To learn Advance Java Softwares Required***
 - ***Eclipse Enterprise Edition***
 - ***MySQL Workbench***
 - ***MySql Server***
 - ***Tomcat V9.0***
 - ***Spring Tool Suite(Optional)***